

**BEFORE YOU READ — ANSWER FROM MEMORY**

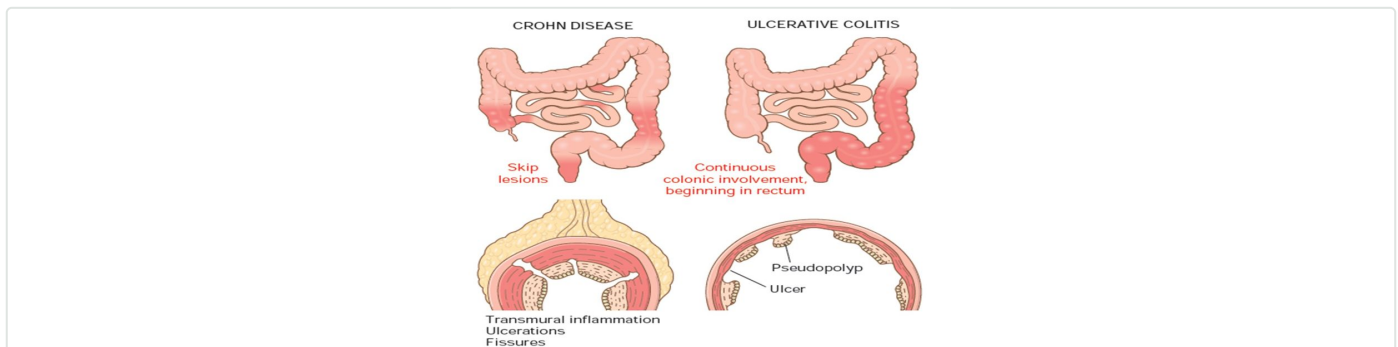
1. Which of Crohn and UC has **skip lesions**, and which is **continuous**?
2. Which one shows **granulomas** on biopsy?
3. In Hirschsprung disease, is the **narrow** or the **dilated** segment the diseased one?

Guessing wrong before reading still improves retention — attempt all three, then check as you go.

1 Inflammatory bowel disease — the frame

Definition	Chronic conditions resulting from inappropriate mucosal immune activation .
Character	Chronic, remitting and relapsing inflammation of unknown etiology .
Two entities	Crohn disease (CD) and ulcerative colitis (UC) .
Pathogenesis	Uncertain — believed to combine altered host reaction to intestinal microbiota , intestinal epithelial dysfunction , aberrant mucosal immune responses , and an altered gut microbiome .
Clinical features	Altered bowel habit · rectal bleeding, urgency and tenesmus · abdominal cramping and pain · nausea and vomiting · systemic: weight loss, fever, sweating, malaise, arthralgia .

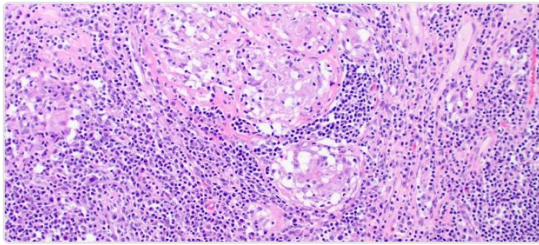
The clinical picture is shared. **The distinction between Crohn and UC is based primarily on morphology** — which is why the table below is the highest-value thing in this lecture.

2 Crohn vs ulcerative colitis — the discriminator

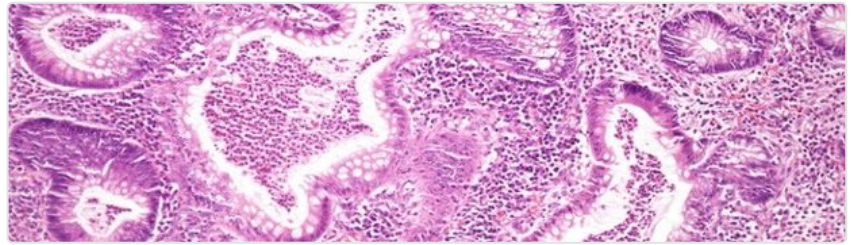
Distribution of lesions: Crohn — skip lesions, transmural inflammation, fissures · UC — continuous colonic involvement beginning in the rectum, ulcers and pseudopolyps

Feature	Crohn disease	Ulcerative colitis
Distribution	Skip lesions — diseased segments with intervening normal mucosa	Continuous , beginning in the rectum , with an abrupt transition to normal mucosa
Depth	Transmural	Mucosal / submucosal
Ulcers	Longitudinal ulcers; fissuring ulceration	Broad-based ulceration
Granulomas	Non-necrotizing granulomas present	Absent
Fibrosis	Marked , with transmural and subserosal lymphoid aggregates	Not a feature
Mucosal surface	Cobblestoning from longitudinal ulcers + normal mucosa	Hyperaemic mucosa with numerous pseudopolyps
Biopsy hallmark	Granuloma, transmural lymphoid aggregates	Crypt abscesses , crypt distortion, mucin depletion , regenerative atypia

If you remember only two things: Crohn **skips** and **goes through the wall** and has **granulomas**; UC is **continuous from the rectum** with **crypt abscesses** and **no granulomas**. Nearly every IBD question hangs on one of those.



Crohn — non-necrotizing granuloma



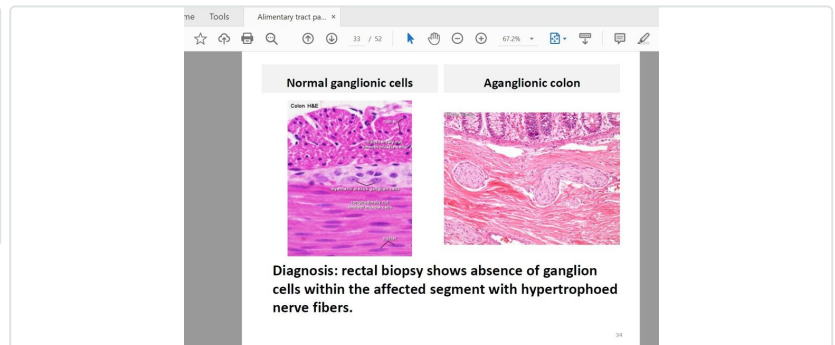
UC — crypt abscesses, crypt distortion, mucin depletion

3 Hirschsprung disease — illness script

WHO gets it	Neonates — about 1 in 5000 live births . Also called congenital aganglionic megacolon .
WHY it happens	Migration of neural crest cells from cecum to rectum is arrested prematurely , or ganglion cells die prematurely. The distal segment then lacks both the Meissner submucosal and Auerbach myenteric plexus — "aganglionosis".
WHAT results	Absence of peristalsis → functional obstruction . The aganglionic segment is narrow ; the normal bowel proximal to it dilates .
WHAT you see	Failure to pass meconium in the immediate postnatal period → abdominal distention and bilious vomiting . Contrast enema: small-calibre affected segment with proximal dilatation.
Diagnosis	Rectal biopsy — absence of ganglion cells in the affected segment, with hypertrophied nerve fibres .
NEVER miss	Complications: enterocolitis , fluid and electrolyte disturbance, perforation and peritonitis . Treatment is surgical resection of the aganglionic segment with anastomosis .



Marked abdominal distention in a neonate



Rectal biopsy — normal ganglion cells vs aganglionic segment

The counter-intuitive bit students lose marks on: the *narrow* segment is the diseased one. The dilated bowel is normal bowel backing up behind it — so biopsy the narrow part, not the obvious swollen part.

P Must-remember pearls

- **Crohn skips; UC is continuous from the rectum**. That single line answers most IBD questions.
- **Granulomas mean Crohn**. Non-necrotizing, and absent in UC.
- **Crohn is transmural** — hence fibrosis, fissures and strictures. UC stays superficial.
- **Crypt abscesses with mucin depletion point to UC**.
- **Hirschsprung: failure to pass meconium** is the presenting sign; the **narrow** segment is the aganglionic one.
- **Hirschsprung lacks BOTH plexuses** — Meissner submucosal and Auerbach myenteric.

Q Test yourself

1. A colectomy specimen shows continuous involvement from the rectum proximally, with an abrupt transition to normal mucosa and numerous pseudopolyps. Which finding would you expect on biopsy?

- A. Non-necrotizing granulomas
- B. Crypt abscesses with mucin depletion
- C. Transmural lymphoid aggregates
- D. Fissuring ulceration

2. A 24-year-old has longitudinal ulcers with intervening normal mucosa and marked mural fibrosis. Which complication follows most directly from the depth of inflammation?

- A. Pseudopolyp formation
- B. Toxic megacolon
- C. Stricture and fistula formation
- D. Rectal bleeding

3. A neonate fails to pass meconium and develops abdominal distention and bilious vomiting. Contrast enema shows a narrow distal segment with proximal dilatation. Where should the biopsy be taken?

- A. From the dilated proximal segment
- B. From the narrow distal segment
- C. From the transition zone only
- D. Biopsy is not required

4. Which single histological finding, if present, most strongly favours Crohn disease over ulcerative colitis?

- A. Crypt distortion
- B. Mucin depletion
- C. Non-necrotizing granuloma
- D. Mucosal ulceration

5. In Hirschsprung disease, the functional obstruction results from:

- A. Mechanical narrowing by fibrous tissue
- B. Absence of peristalsis in the aganglionic segment
- C. Hypertrophy of the muscularis propria
- D. Meconium impaction in the ileum

ANSWERS — COVER UNTIL YOU'VE COMMITTED

1 • B — Continuous rectal-onset disease with pseudopolyps is ulcerative colitis; crypt abscesses with mucin depletion are its biopsy hallmark. The other three options are all Crohn features.

2 • C — Crohn is transmural, so inflammation and fibrosis reach through the wall — producing strictures and fistulae. UC's superficial disease does not.

3 • B — The narrow segment is the aganglionic diseased bowel; the dilated part is normal bowel obstructed behind it. Biopsying the dilated segment would show normal ganglion cells.

4 • C — Non-necrotizing granulomas are present in Crohn and absent in UC. Crypt distortion, mucin depletion and ulceration occur in both.

5 • B — Aganglionosis removes both the Meissner and Auerbach plexuses, so peristalsis fails — a functional, not mechanical, obstruction.

SPACED REVIEW

☐ Day 1

☐ Day 3

☐ Day 7

☐ Day 21

☐ Pre-exam

Pathology GIT Block · MEDucation by Mattin Majid · summarised from LG3, Dr Jawher Tahir Omer, Hawler Medical University
Study alongside the lecture, not instead of it.